Software Test Plan (STP) - Collaborative Study Material & Quiz Platform

Project: Collaborative Study Material & Quiz Platform

Current editable source: TEST_PLAN.md. TEST_PLAN.pdf is regenerated from this source. The original Word/PDF plan is preserved under docs/historical with unchanged content. This update changes scheduling and CI terminology, not approved requirements or recorded test outcomes.

Regeneration: run `python docs/tools/generate_test_plan.py` with ReportLab available. The Markdown is the maintained editable planning source; the archived Word original is superseded.

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Authors: Team 10

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Status: Planning update for review; approval signatures are not asserted

1. Introduction

Purpose: This document defines the test plan for the Collaborative Study Material & Quiz Platform v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers authentication, study materials, question bank, quiz creation/search, quiz attempt/evaluation, results/history, subject-wise performance, administration, security, performance, reliability, compatibility, and usability.

References: Final two-sprint plan, current work division, Collaborative Study Material & Quiz Platform SRS v1.5, Detailed Work Division for a 4-Member Software Engineering Team, UML specifications.

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), UI (User Interface), API (Application Programming Interface), MCQ (Multiple Choice Question), RTM (Requirements Traceability Matrix).

2. Test Items

- Authentication module
- Study Material Management module
- Question Bank Management module
- Quiz Creation and Search module
- Quiz Attempt and Evaluation module
- Results, History and Performance module
- Administrator/Admin and Security interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

- CSMQ-F-001, F-002, F-004: Registration, login and logout
- CSMQ-F-010, F-012, F-013, F-041: Study-material upload, search/view, download and admin removal
- CSMQ-F-020, F-022, F-042: Question-bank add, search/filter and admin question/quiz removal
- CSMQ-F-030, F-032: Quiz creation and search/view
- CSMQ-F-033, F-034, F-035, F-036: Quiz attempt, evaluation, results and history
- CSMQ-F-040: Subject-wise average score
- CSMQ-NF-001: Response time within 3s
- CSMQ-NF-002: Data persistence after restart

- CSMQ-NF-003: At least 10 concurrent sessions

- CSMQ-NF-004: Supported browser compatibility

- CSMQ-NF-005: Clear and understandable UI feedback

4. Features Not to be Tested

- Enterprise/distributed-scale infrastructure beyond the single local/development deployment

- Specialized hardware interfaces (none are required by the SRS)

- Functionality not defined in the approved SRS

5. Test Approach and Strategy

Levels:

- Unit tests (module-level)

- Integration tests (frontend, REST APIs, backend, MongoDB)

- System tests (end-to-end Student and Administrator functionality)

- Acceptance tests (UAT)

Types:

- Functional testing (core features)

- Regression testing

- Performance testing (response time, concurrent users)

- Usability testing (UI clarity, navigation, feedback)

Entry Criteria: Stable build delivered, test data available, services and test environment ready.

Exit Criteria: 100% of planned test cases executed, High-priority requirements verified, no unresolved critical defects, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password hashing and confirm no plain-text password storage

- Verify authentication before protected pages, functions and APIs

- Verify role-based access control for Administrator-only operations

- Validate and safely handle malformed or malicious input

- Verify HTTPS for sensitive communication in the deployed environment

6. Test Environment

Hardware: Desktop/laptop/tablet/smartphone devices; no specialized hardware required.

Software: React.js frontend, Node.js, Express.js REST API, MongoDB, modern web browsers.

Tools: Postman (API), browser developer tools, Jira (defect tracking), GitHub (test evidence/PRs).

Test Data: Student and Administrator accounts, valid/invalid credentials, PDF files, MCQs, quizzes, attempts and results.

7. Test Schedule

The project has exactly two implementation sprints. Planning and setup are activities within the first sprint, not a separate implementation sprint.

Sprint	Test scope	Completion evidence
Sprint 1	Core authentication, PDF upload and validation, Question Bank, QuizAttempt/evaluation prototype, security/RBAC foundation, unit/integration testing and GitHub Actions CI/CD	Working first increment; tests/coverage; genuine reviews; PR history; Jira evidence; Sprint 1 demonstration
Sprint 2	Remaining study-material features, complete quiz creation/search and submission/evaluation/results/history/performance, administrator functions, integration, security, persistence, regression/system/NFR testing, performance/concurrency, browser/usability, cross-testing, UAT, deployment/demo and release readiness	All approved SRS acceptance criteria verified on the full integrated application; defects resolved or formally accepted; final reports, Jira/GitHub evidence, final demo and release-ready build

Unit and integration tests progress alongside development in both sprints. All remaining product, validation and release work must finish within Sprint 2; the schedule does not declare that this work is already implemented. GitHub Actions is the active build/test pipeline.

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts / API test collections
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Harshit Chandak	Prepare plan, coordinate execution
Test Engineer	Team 10	Design & execute test cases, log defects
Developer	Team 10	Support defect fixes and triage
Product Owner	Project Guide	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from development team
Test environment or service downtime	Maintain backup test setup and restore procedures
Integration/API changes cause retesting delays	Coordinate contracts early and run regression smoke tests

11. Assumptions & Dependencies

- MongoDB test database and application services will be stable and available
- Required Student/Administrator test accounts and sample data will be available before execution

- The integrated MERN build and required browser versions will be available for testing

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment is unavailable for a material portion of the test window

- Build instability blocks more than 30% of planned test cases

Resume testing if:

- Blocking defects are resolved or formally accepted

- Environment is stabilized and smoke tests pass

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.

Example:

- CSMQ-F-001 (Registration) -> TC-Auth-01

- CSMQ-F-030 (Quiz creation) -> TC-QUIZ-01

- CSMQ-NF-001 (Response time) -> TC-Perf-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed

- % passed/failed

- Defect density

- Defect aging

- Requirement coverage

Reports:

- Daily execution status

- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Harshit Chandak	
Dev Lead	Team 10	
Product Owner	Project Guide	